

Exact Two-Clock Periodic Geometry of Cyclic Rule 90

Route-A structural certificate C145

Abstract

We count every temporally periodic configuration of Rule 90 on every finite cyclic binary lattice. For spatial circumference L and time n , the fixed count is a power of two whose exponent is an explicit polynomial-gcd degree. An ideal-theoretic proof covers even L , where the spatial polynomial can be non-squarefree. Möbius inversion then separates exact-period points from primitive temporal cycles, and fixed configurations are exactly labeled $L \times n$ spatiotemporal tori. A complete 24×24 ledger gives domain-qualified aspect-ratio witnesses and a same-fixed-count control with different primitive content. The result is exact finite-volume progress, not a single frozen target determinant.

1 Frozen Rule-90 family

On the cyclic lattice $\mathbb{Z}/L\mathbb{Z}$, Rule 90 is

$$(F_L u)_i = u_{i-1} + u_{i+1} \pmod{2}.$$

Both summands are retained when the two neighbor indices coincide; over $\mathbb{F}_2$ they then cancel. Thus the definition also covers $L = 1, 2$. Encode u in $R_L = \mathbb{F}_2[x, x^{-1}]/(x^L - 1)$; then F_L is multiplication by $a = x + x^{-1}$. Freeze the ordered pair (L, n) as the spatial and temporal clocks. Write $\text{Fix}(L, n) = \#\ker(F_L^n - I)$, $P_L(n)$ for points of least positive temporal period n , and $C_L(n)$ for their geometric temporal cycles. These three quantities are not interchangeable.

2 Kernel-gcd theorem

Multiplication lemma. For monic $f \in k[x]$ and any h , multiplication m_h on $k[x]/(f)$ satisfies

$$\dim \ker m_h = \deg \gcd(f, h). \tag{1}$$

Indeed, put $g = \gcd(f, h)$, $f = gf_1$, and $h = gh_1$. Since $\gcd(f_1, h_1) = 1$,

$$f \mid hq \iff f_1 \mid q.$$

The kernel is therefore represented uniquely by $f_1 r$ with $\deg r < \deg g$, which proves (1): these representatives span, and $f \mid f_1(r - r')$ would force $g \mid (r - r')$, hence $r = r'$ by the degree bound. This also covers $\deg g = 0$. The argument uses divisibility rather than distinct roots.

Because x is invertible modulo $x^L + 1 = x^L - 1$ in characteristic two, clearing the Laurent denominator does not change the kernel:

$$x^n(a^n - 1) = (x^2 + 1)^n + x^n =: b_n. \tag{2}$$

Applying (1) gives the all-size formula

$$\boxed{\text{Fix}(L, n) = 2^{\deg \gcd(x^L + 1, (x^2 + 1)^n + x^n)}}, \quad L, n \geq 1. \tag{3}$$

No squarefree hypothesis occurs: for example $x^6 + 1 = (x^3 + 1)^2$ over $\mathbb{F}_2$, and (3) still applies.

3 Exact temporal cycles and tori

Even when F_L is not invertible on the whole state space, a point satisfying $F_L^n u = u$ lies on a cycle whose least period divides n . Hence the fixed-point set at time n is the disjoint union of exact-period sets at divisors of n . Möbius inversion and orbit partitioning give

$$P_L(n) = \sum_{d \mid n} \mu(n/d) \text{Fix}(L, d), \quad C_L(n) = \frac{P_L(n)}{n} \in \mathbb{Z}_{\geq 0}. \tag{4}$$

The division occurs only after inversion: exact-period- n points form cycles of size n .

A labeled spatiotemporal torus is an array satisfying

$$u_{i,j+1} = u_{i-1,j} + u_{i+1,j}, \quad i \bmod L, \quad j \bmod n.$$

Its row at $j = 0$ determines all rows, and temporal closure is precisely $F_L^n u = u$. Thus the number of labeled $L \times n$ tori is $\text{Fix}(L, n)$.

4 Two-clock witnesses

The table records exact counts for $1 \leq L, n \leq 24$. Every minimum below is within that frozen cutoff, not over unbounded L, n ; the bounded-search domains are distinct:

search subset	area	cells $(L, n; \text{Fix})$	nonzero exact content
$1 \leq L, n \leq 24$	3	$(1, 3; 1), (3, 1; 4)$	$(3, 1) : P = 4$
$2 \leq L, n \leq 24$	6	$(2, 3; 1), (3, 2; 4)$	none at displayed times
$2 \leq L, n \leq 24$, some $P > 0$	12	$(2, 6; 1), (3, 4; 4), (4, 3; 1), (6, 2; 16)$	$(6, 2) : P = 12, C = 6$

For each area the search enumerates every eligible divisor pair (L, n) in the stated box; no best-case pair is omitted. Thus area Ln loses aspect ratio. A different control fixes $L = 5$:

$$\text{Fix}(5, 3) = \text{Fix}(5, 6) = 16, \quad (P_5(3), C_5(3)) = (15, 5), \quad (P_5(6), C_5(6)) = (0, 0). \quad (5)$$

One fixed count therefore loses the divisor history that determines primitive content.

5 Validation and Route-A boundary

Across 576 cells, the ledger sums to 488,334 fixed points, 283,758 exact-period points, and 24,474 primitive cycles. The independent binary-matrix checker passes 6,520 assertions, including direct state enumeration at small sizes. SymPy passes 1,177 exact polynomial checks, production replays byte-identically, and all 43 hostile cases (42 repaired-hash and one stale-hash) are rejected.

The strict tuple is `(A1_WEAK, A2_FAIL, A3_FAIL, A4_FAIL)`, overall `ROUTE_A_EXPLORATORY`. The intrinsic finite-volume cycles depend on two essential clocks; we supply no single frozen target divisor, target functional equation or counting law, arithmetic/local factor, root number, automorphy, natural operator lift, Hilbert–Pólya operator, infinite-volume limit, or Route-B authorization. The scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`.